

Ontology for Laptops

(with Reasoner and DL querier)

Sneha Oram, IIT Bombay

08th May 2026

Problem Statement

To create ontology for a product (laptop) and integrate a Reasoner to check the consistency of the ontology. Subsequently, use of DL queries for inferencing.

Ontology design

The idea is to create a hierarchy of classes for specifications of laptops such as display (screen), memory (RAM), operating system (OS), processor (CPU), storage (hard drive/SSD).

Experiment setup

Platforms: Desktop Protégé (Version: 5.6.9) / Web Protégé (by (BMIR) at Stanford University)

Desktop Protégé Plugins:

- DL Query (Version: 4.0.1)
- HermiT Reasoner (Version: 1.4.3)

Terminologies in Ontology

- IRI (Internationalized Resource Identifier): A unique identifier used to represent ontology elements (ontology and entities) on the web.
- Active ontology: The ontology currently loaded and being edited or analyzed.
- Entities: The main components used to define knowledge in an ontology.
 - Class: Represents a category or group of similar objects or concepts.
 - Object Properties: Define relationships between two individuals or entities.
 - Data Properties: Define relationships between individuals and data values.
 - Individuals: Specific instances or examples of a class.

Entities: Classes

Following classes are created for developing ontology for laptops. The class hierarchy is presented on the right.

- Device
 - Computers
 - Laptops
- Components
 - Processor (CPU)
 - Memory (RAM)
 - Storage (Hard Drive/SSD)
 - Display (Screen)
 - OperatingSystem (OS)

All classes are subclass of owl:Thing (superclass of all classes)

Entities: Object Properties (Relationships)

This links the Laptops to their components

- Component
 - hasComponent (Domain: Device, Range: Component)
 - hasDisplay (Sub-property of hasComponent)
 - hasMemory
 - hasOS
 - hasProcessor
 - hasStorage

All object properties are subclass of the owl:topObjectProperty

Entities: Data Properties (Attributes)

These hold the literal values (numbers, strings) for the specifications

- modelName (Range: string)
- osName (Range: string)
- processorName (Range: string)
- ramSizeGB (Range: integer)
- storageCapacityGB (Range: integer)
- screenSizeInches (Range: float/decimal)

All data properties are subclass of the owl:topDataProperty

Entities: Individuals (The Laptops) [1/5]

This is to create instances for the specific hardware components first, then the laptops themselves. Following are the component individuals that are created with the details of the data properties assigned:

- Instance: Intel_Core_i5_1135G7 (Class: Processor)
 - processorName: Intel Core i5-1135G7
- Instance: Apple_M2_Chip (Class: Processor)
 - processorName: Apple M2 (8-core CPU)
- Instance: Dell_RAM_16GB (Class: Memory)
 - ramSizeGB: 16

Entities: Individuals (The Laptops) [2/5]

- Instance: Apple_Unified_Memory_16GB (Class: Memory)
 - ramSizeGB: 16
- Instance: Dell_Screen_15.6 (Class: Display)
 - screenSizeInches: 15.6
- Instance: Mac_Liquid_Retina_13.6 (Class: Display)
 - screenSizeInches: 13.6
- Instance: Windows_11_Pro (Class: OperatingSystem)
 - osName: Windows 11 Pro

Entities: Individuals (The Laptops) [3/5]

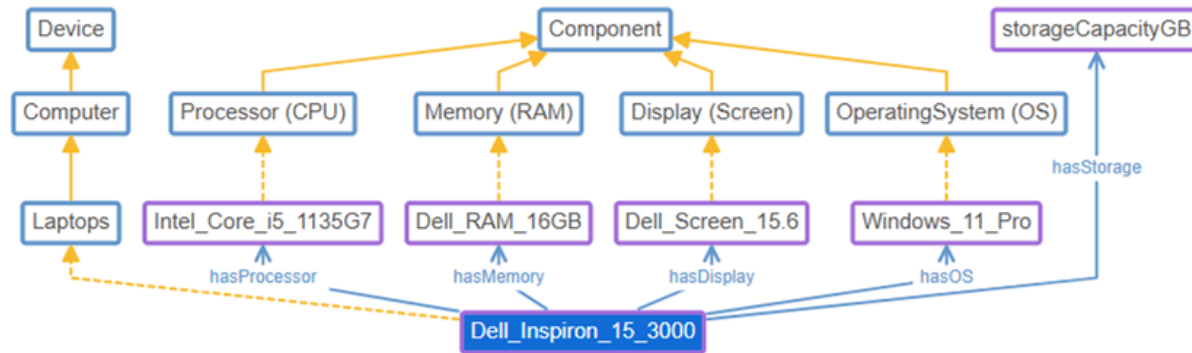
- Instance: macOS_Monterey (Class: OperatingSystem)
 - osName: macOS Monterey

Entities: Individuals (The Laptops) [4/5]

Next, Laptop Individuals are created. Main Laptop instances are created first, and then linked to the components using object properties. Below are the details of Dell Inspiron 15 3000

- Individual: Dell_Inspiron_15_3000
 - Class: Laptop
 - Data Property Assertions:
 - modelName: Dell Inspiron 15 3000
 - Object Property Assertions
 - hasProcessor -> Intel_Core_i5_1135G7 ■ hasOS -> Windows_11_Pro
 - hasMemory -> Dell_RAM_16GB
 - hasDisplay -> Dell_Screen_15.6

- Individual: Dell Inspiron 15 3000

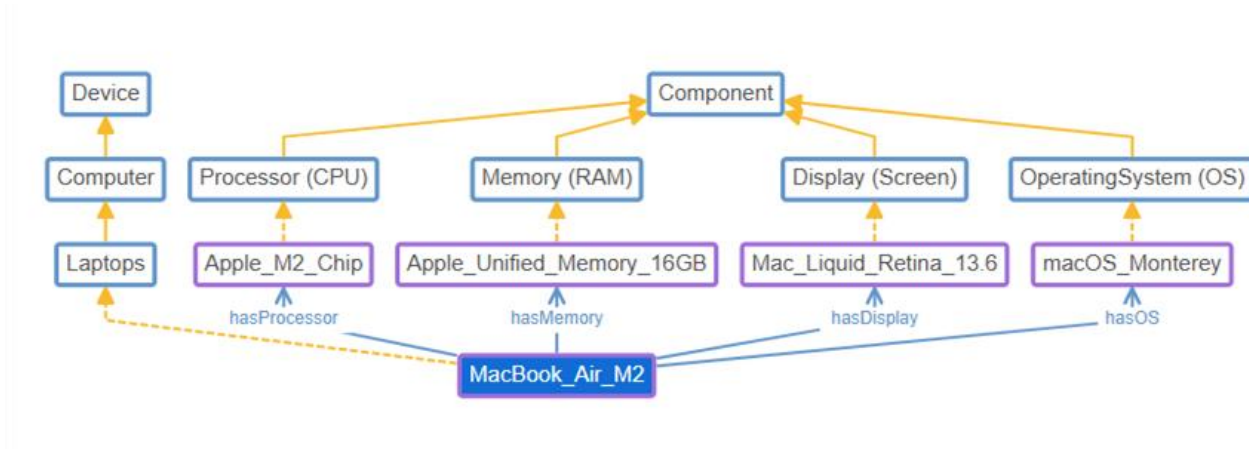


Entities: Individuals (The Laptops) [5/5]

Below are the details of MacBook Air M2

- Individual: MacBook_Air_M2
 - Class: Laptop
 - Data Property Assertions:
 - modelName: MacBook Air M2
 - Object Property Assertions
 - hasProcessor -> Apple_M2_Chip
 - hasMemory -> Apple_Unified_Memory_16GB
 - hasDisplay -> Mac_Liquid_Retina_13.6
 - hasOS -> macOS_Monterey

- Individual: MacBook Air M2



HermiT Reasoner

- Reasoner (HermiT (selected by default))
 - Start Reasoner: This starts verifying the individuals with respect to their relationships and data properties. In case of any error, the reasoner presents an option 'Explain', where the inconsistencies with constraints or logic definition are pointed.

DL queries

- Example queries and results

Query (class expression)

Laptops

Execute Add to ontology

Query results

Subclasses (1 of 1)

● owl:Nothing

Instances (3 of 3)

◆ Dell_Inspiron_15_3000

◆ MacBook_Air_M2

Query (class expression)

Laptops and hasMemory some

Execute Add to ontology

Query results

Subclasses (1 of 1)

● owl:Nothing

Instances (2 of 2)

◆ Dell_Inspiron_15_3000

◆ MacBook_Air_M2

Query (class expression)

hasProcessor some

Execute Add to ontology

Query results

Subclasses (1 of 1)

● owl:Nothing

Instances (2 of 2)

◆ Dell_Inspiron_15_3000

◆ MacBook_Air_M2

Query (class expression)

hasOS value macOS_Monterey

Execute Add to ontology

Query results

Subclasses (1 of 1)

● owl:Nothing

Instances (1 of 1)

◆ MacBook_Air_M2

Query (class expression)

Laptops and hasProcessor value Intel_Core_i5_1135G7

Execute Add to ontology

Query results

Subclasses (1 of 1)

● owl:Nothing

Instances (1 of 1)

◆ Dell_Inspiron_15_3000

Query (class expression)

Laptops and hasMemory some (ramSizeGB value 16)

Execute Add to ontology

Query results

Subclasses (1 of 1)

● owl:Nothing

Instances (2 of 2)

◆ Dell_Inspiron_15_3000

◆ MacBook_Air_M2

Notes

- This experiment is conducted on simple ontology for laptops.
- In future, the ontology will be expanded with other specifications. Further, constraints and logics to be validated by the Reasoner.

Thank You